

Influence_Consensus_agr_een - In society, does Agriculture alone generally have a direct effect on Energy Production (green energy, fuel, oil, natural gas, cost of utilities and production, etc.), or are observed relationships primarily indirect? If a direct effect exists, is it negative, none, negligible, or positive? What would be a reasonable normalized magnitude for this influence on a scale from -1.00 to 1.00? Please only provide evidence for only direct influences

Agriculture has a direct and multifaceted effect on energy production and consumption, primarily through its role as both a significant energy consumer and a renewable energy producer. Agricultural activities consume large amounts of fossil fuels, especially for fertilizers, machinery, and irrigation, with high-input farming practices increasing fossil fuel energy demand by about five times compared to low-input systems (Rosa et al., 2021; Woods et al., 2010). However, agriculture also directly contributes to renewable energy production through biomass, biogas, and biomethane generation, which can decentralize energy supply and reduce reliance on external fossil fuels (Sulewski & Wąs, 2024; Švažas & Navickas, 2025; Pestisha et al., 2023). The productivity of direct energy inputs in agriculture is influenced by mechanization and production intensity, which improve energy efficiency, while larger scale and animal farming tend to reduce it (Wicki et al., 2025). Despite improvements in energy efficiency, the overall direct impact of agriculture on energy use is complex due to rebound effects where increased efficiency can lead to higher total consumption (Han & Zhou, 2023). On balance, the direct influence of agriculture on energy production and consumption is positive but moderate, reflecting both its role as an energy user and a growing renewable energy prosumer; a reasonable normalized magnitude would be around +0.3 on a scale from -1.00 to 1.00.

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References

- Han, H., & Zhou, Z. (2023). The rebound effect of energy consumption and its determinants in China's agricultural production. *Energy*. <https://doi.org/10.1016/j.energy.2023.129961>
- Pestisha, A., Gabnai, Z., Chalgybayeva, A., Lengyel, P., & Bai, A. (2023). On-Farm Renewable Energy Systems: A Systematic Review. *Energies*. <https://doi.org/10.3390/en16020862>
- Rosa, L., Rulli, M., Ali, S., Chiarelli, D., Dell'Angelo, J., Mueller, N., Scheidel, A., Siciliano, G., & D'Odorico, P. (2021). Energy implications of the 21st century agrarian transition. *Nature Communications*, 12. <https://doi.org/10.1038/s41467-021-22581-7>
- Sulewski, P., & Wąs, A. (2024). Agriculture as Energy Prosumer: Review of Problems, Challenges, and Opportunities. *Energies*. <https://doi.org/10.3390/en17246447>
- Švažas, M., & Navickas, V. (2025). The Synergy Potential of Energy and Agriculture—The Main Directions of Development. *Energies*. <https://doi.org/10.3390/en18051031>
- Wicki, L., Dudek, H., Parzonko, A., Kusz, D., & Naglis-Liepa, K. (2025). Factors Influencing the Productivity of Direct Energy Inputs in EU Agriculture. *Sustainability*. <https://doi.org/10.3390/su17031217>
- Woods, J., Williams, A., Hughes, J., Black, M., & Murphy, R. (2010). Energy and the food system. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 365, 2991 - 3006. <https://doi.org/10.1098/rstb.2010.0172>